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/ **Practice Exam - 1**

Correct Answer

Next C

Practice Exam - 1

Question 25 of 117



A 48-year old with longstanding but well-controlled human immunodeficiency virus infection presents with progressive dyspnea on exertion.

Vitals signs are notable for blood pressure 100/70 mm Hg, heart rate 90 beats per minute, respiratory rate 12 breaths per minute with oxygen saturation 91% on room air. Cardiac examination reveals a right ventricular heave, regular rate and rhythm with prominent P2 and a systolic murmur at the left lower sternal border.

Transthoracic echocardiogram reveals left ventricular ejection fraction of 65%, moderately reduced right ventricular systolic function with moderately dilated right ventricle. There is moderate tricuspid regurgitation. Estimated right ventricular systolic pressure is 45 mm Hg.

Subsequently, a right heart catheterization is performed and reveals the following: right atrial pressure 14 mm Hg, pulmonary artery pressure 84/42 mm Hg with mean PA 56 mm Hg, pulmonary capillary wedge pressure 13 mm Hg. Cardiac output 4 L/min with pulmonary vascular resistance 11 Wood units.

Inhaled nitric oxide (NO) is administered at escalating doses from 20 PPM to 80 PPM.

Which of the following hemodynamic responses to NO suggest the patient would benefit from treatment with calcium channel blockers?

A. Mean pulmonary artery pressure of 50 mm Hg with cardiac output of 3L/min

B. Pulmonary capillary wedge pressure of 15 mm Hg with cardiac output of 4L/min

C. Mean pulmonary artery pressure of 35 mm Hg with cardiac output of 4L/min

D. Pulmonary vascular resistance of 8 Wood units with cardiac output of 5L/min

Testing Point: The Board-certified AHFTC specialist should know what constitutes a positive response to vasodilator testing for pulmonary arterial hypertension.

Answer: C. Mean pulmonary artery pressure of 35 mm Hg with cardiac output of 4L/min.

A positive response to pulmonary vasodilator testing should include all of the following: 1) decrease in mPAP to ≤ 40 mm Hg; 2) decrease of at least 10 mm Hg in mPAP; and 3) unchanged or increased cardiac output. In patients with a positive response, treatment with calcium channel blockers should be considered as a first-line agent.

Answer D: While such a response typically results when there is a drop in PVR, current definitions of acute vasodilator response do not include PVR.

Answer B: Wedge pressure is not included in the definition of pulmonary responsiveness, although a low wedge pressure is an essential part of the diagnosis of pulmonary arterial hypertension.

Answer A shows an insufficient drop in PA pressure with a fall in cardiac output and thus does not meet the definition of responsiveness.

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